

SEMESTER END EXAMINATIONS - FEBRUARY 2024

Program : **B.E :- Information Science and Engineering**
Course Name : **Theory of Computation**
Course Code : **IS54**

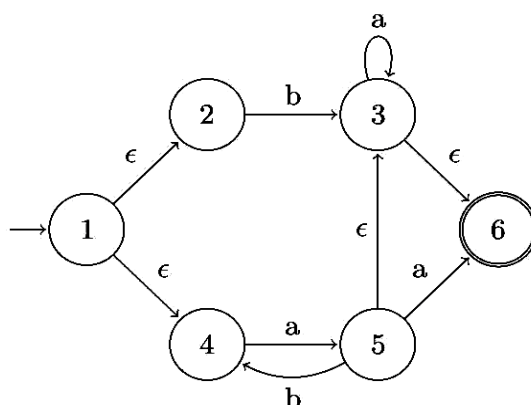
Semester : **V**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

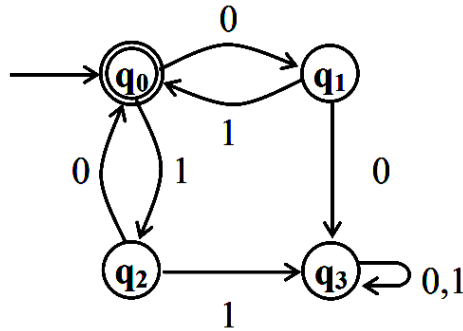
- Construct a Deterministic Finite Automata(DFA), which accepts set of all strings over $\{0,1\}$, which when interpreted binary number is divisible by 5. CO1 (08)
 - Construct a DFA for all the strings over $\{a,b\}$ which contain exactly 2a's and exactly 3b's. CO1 (08)
 - Construct a DFA to accept the language $L = \{w \in \{a, b, c\}^* | w \text{ starts and ends with the same symbol}\}$.
 - Show by giving an example that, if M is an Non-deterministic Finite Automata(NFA) that recognizes language L, swapping the accept and non-accept states in M doesn't necessarily yield a new NFA that recognize L' (compliment of L). CO1 (04)
- Define Non-Deterministic Finite State Automata (NFA). Construct a NFA to accept the following languages: CO1 (08)
 - $L = \{w \in \{a,b\}^* : \exists x, y \in \{a,b\}^* ((w=xabbaay) \vee (w=xbabay))\}$
 - $L = \{w \in \{a,b\}^* : w \text{ is made up of an optional 'a' followed by 'aa' followed by zero or more b's}\}$
 - Discuss the following with examples: CO1 (04)
 - Kleene Star of Language
 - Concatenation of Languages.
 - Convert the given epsilon-NFA to Deterministic Finite Automata(DFA). CO1 (08)



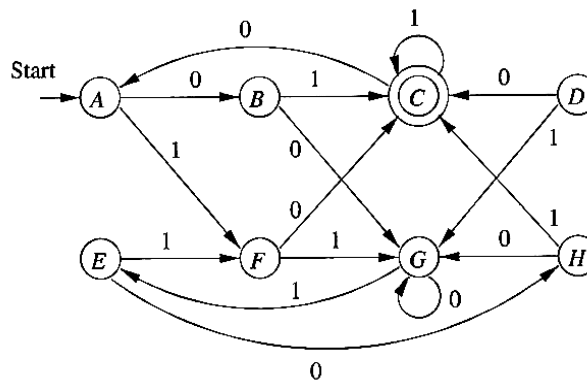
UNIT - II

- Write down the regular expressions for the following languages: CO2 (08)
 - Strings of a's & b's containing not more than three a's.
 - $L = \{a^{2n}b^{2m} \mid n \geq 0, m \geq 0\}$
 - To accept a language consisting of string of a's and b's of even length.
 - Set of strings of 0's and 1's whose tenth symbol from the right end is 1.

- b) Using state elimination method obtain a regular expression for the FA CO2 (06)
shown below:



- c) Illustrate with examples the precedence of operators used in regular CO2 (06)
expression.
4. a) State and prove that if $L = L(A)$ for some DFA A, then there is a regular CO2 (10)
expression R such that $L = L(R)$.
- b) Consider the diagram: CO2 (10)



- i. Draw the table of distinguish abilities for this automaton.
ii. Construct the minimum state equivalent DFA.

UNIT - III

5. a) Obtain CFG for the given languages: CO3 (08)
i. $L = \{WW^R \mid W \in \{a,b\}^*\}$
ii. $L = \{n_a(W) = n_b(W) \mid W \in \{a,b\}^*\}$
iii. $L = \{W \in \{ (,), [,] \}^* \mid W \text{ contains balanced parentheses} \}$
iv. $L = \{a^n b^n \mid n \geq 0\}$
- b) What is ambiguous grammar? Show that the following grammar is CO3 (12)
ambiguous:
 $S \rightarrow 0B \mid 1A$
 $A \rightarrow 0S \mid 1AA \mid 0$
 $B \rightarrow 1S \mid 0BB \mid 1$
6. a) Obtain CNF for the given Grammar: CO3 (08)
 $S \rightarrow aAb \mid abBB$
 $A \rightarrow aAA \mid BaB$
 $B \rightarrow AAb \mid a$
 $C \rightarrow Saa \mid AB$
- b) What are useless symbols? Write the procedure to eliminate useless CO3 (08)
symbol from CFG.
- c) Discuss any two applications of CFG. CO3 (04)

UNIT- IV

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| 7. | a) | Design a PDA to accept the set of all strings over $\{a, b\}$ with equal number of a's and b's. Show the moves for abbaba. | CO4 | (10) |
| | b) | Describe with a neat diagram the working principles of PDA. | CO4 | (06) |
| | c) | Describe the information conveyed by Instantaneous description (ID) in PDA. | CO4 | (04) |
| 8. | a) | Describe language of PDA accepted by final state and empty stack. | CO4 | (05) |
| | b) | Differentiate between DPDA and NDPA. | CO4 | (05) |
| | c) | Convert the following grammar to a PDA that accept the same language by empty stack.
$S \rightarrow 0S1 \mid A$
$A \rightarrow 1A0 \mid S \mid \epsilon$ | CO4 | (10) |

UNIT - V

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| 9. | a) | Examine the different programming techniques for turing machines. | CO5 | (10) |
| | b) | Design a Turing machine that accepts $L = \{a^n b^n c^n : n \geq 1\}$. Give the computational Sequence for the input aabbcc. | CO5 | (10) |
| 10. | a) | Differentiate between NP-hard and NP-complete problems. | CO5 | (05) |
| | b) | Explain the working of a standard turing machine and design a TM that accepts the Language $L = \{a^n b^{2n} : n \geq 0\}$. | CO5 | (10) |
| | c) | Explain the extensions to the basic turing machine. | CO5 | (05) |
